

Ontological Collapse: What the Classical Atom Reveals About Scientific Models and the Metaphysics They Carry

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July 3, 2026

Draft for discussion — do not circulate

De paseos, perogrulladas y discusiones varias

Abstract

We present a discussion on the classical calculation of the radiative collapse of the hydrogen atom, and how it constitutes not merely an empirical failure but an *ontological* one. The planetary model of the atom fails not because its equations are internally inconsistent, but because the type of entities it presupposes—such as a point charge with a definite trajectory—does not correspond to what electrons are. We use this case to distinguish two layers present in any physical model, often ignored or not sufficiently emphasized: its predictive mathematical content and its implicit ontological commitments. We then argue that Aristotle’s distinction between potentiality (δύναμις) and actuality (ἐνέργεια), recovered by Heisenberg in 1958, provides a more adequate ontological framework for the quantum atom than either naive particle realism or strict instrumentalism. The paper closes by situating this reading within contemporary debates on structural realism and the metaphysics of quantum objects.

1 The Classical Collapse: An Explicit Calculation

The failure of the planetary models of the atom is a well known and extensively discussed example. We present the calculation in full because the philosophical argument that follows depends on it: the failure is not a rough estimate but an exact consequence of three internally consistent and independently verified physical laws.

1.1 The Three Ingredients and Their Sources

Centripetal acceleration (Newton, 1687)

Newton's *Principia Mathematica* [1] establishes that for uniform circular motion of radius r and speed v , differentiating the position vector $\vec{r}(t) = r(\cos \omega t, \sin \omega t)$ twice with $\omega = v/r$ gives

$$\ddot{\vec{r}} = -\omega^2 \vec{r} \implies \boxed{a = \frac{v^2}{r}}, \quad (1)$$

directed toward the centre. No approximation is involved.

Coulomb's law (Coulomb, 1785)

Measured by torsion balance in 1785 [2], the force arising between two point charges q_1 and q_2 separated by distance r is

$$\boxed{F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}}, \quad (2)$$

attractive for opposite charges. The constant ϵ_0 would be added by Maxwell's 1865; the inverse-square dependence is Coulomb's direct experimental result.

Larmor radiation formula (Larmor, 1897)

From Maxwell's equations, the electric field radiated by an accelerated charge falls as $1/r$, so the Poynting flux integrated over a distant sphere yields a finite radiated power [3]:

$$\boxed{P = \frac{e^2 a^2}{6\pi\epsilon_0 c^3}}. \quad (3)$$

The dependence on a^2 follows because the radiated field is linear in a , and power is quadratic in the field amplitude.

1.2 The Differential Equation for $r(t)$

Setting Coulomb's force equal to the centripetal requirement for the electron (mass m_e , charge e) one obtains:

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{m_e v^2}{r} \implies v^2 = \frac{e^2}{4\pi\epsilon_0 m_e r}. \quad (4)$$

The total mechanical energy is then:

$$E = \frac{1}{2} m_e v^2 - \frac{e^2}{4\pi\epsilon_0 r} = -\frac{e^2}{8\pi\epsilon_0 r}, \quad (5)$$

where the virial theorem ensures $E_{\text{kin}} = -E/2$.

As the electron loses energy by radiation, E decreases and r decreases monotonically. Substituting $a = e^2/(4\pi\epsilon_0 m_e r^2)$ into (3):

$$P = \frac{e^6}{12\pi^3 \epsilon_0^3 m_e^2 c^3 r^4}. \quad (6)$$

Equating total energy $\dot{E} = -P$ and differentiating (5) one obtains:

$$\frac{e^2}{8\pi\epsilon_0 r^2} \dot{r} = -\frac{e^6}{12\pi^3\epsilon_0^3 m_e^2 c^3 r^4} . \quad (7)$$

Rearranging:

$$r^2 dr = -\frac{2e^4}{3(4\pi\epsilon_0)^2 m_e^2 c^3} dt \equiv -C dt . \quad (8)$$

1.3 Integration and Result

Integrating (8) from $r_0 = a_0$ (the Bohr radius, $a_0 \approx 0.529 \text{ \AA}$) to $r = 0$:

$$\int_{a_0}^0 r^2 dr = -C t_{\text{collapse}} \implies \frac{a_0^3}{3} = C t_{\text{collapse}}, \quad (9)$$

giving the final results

$$t_{\text{collapse}} = \frac{(4\pi\epsilon_0)^2 m_e^2 c^3 a_0^3}{4 e^4} \approx 1.6 \times 10^{-11} \text{ s} . \quad (10)$$

Sixteen picoseconds. This is not an order-of-magnitude estimate; it is the exact result of integrating three independently verified and internally consistent physical laws. The calculation is impeccable, yet the conclusion is empirically false. That tension is the starting point of everything that follows.

Constant	Symbol	Value
Bohr radius	a_0	$0.529 \times 10^{-10} \text{ m}$
Electron mass	m_e	$9.109 \times 10^{-31} \text{ kg}$
Elementary charge	e	$1.602 \times 10^{-19} \text{ C}$
Speed of light	c	$2.998 \times 10^8 \text{ m s}^{-1}$
Permittivity of vacuum	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$

Table 1: Physical constants used in the collapse calculation.

2 Two Layers in Every Physical Model

The collapse calculation exemplifies a general structure present in all physical models, but rarely made explicit. Every model carries at least two distinguishable layers of content:

1. **A predictive-mathematical layer:** the equations, boundary conditions, and inference rules that generate quantitative predictions about measurable quantities.
2. **An ontological layer:** the implicit commitments about what kinds of entities exist, what properties they possess independently of measurement, and how they relate causally.

The atom is our primary case study, but the diagnostic generalises: we identify the same two-layer structure in the ultraviolet catastrophe and in the quantum measurement problem. Standard empiricist philosophy of science focuses almost exclusively on the first layer [4]. A model is adequate if its predictions agree with experiment; inadequate otherwise. On this view, the classical atomic model fails because it predicts a collapse timescale of $\sim 10^{-11}$ s while atoms are stable indefinitely. That diagnosis is correct, but incomplete.

2.1 The Ontological Layer of the Classical Model

The deeper diagnosis is that the classical model fails at the second layer. Newton’s equations for dynamics, Coulomb’s law for electromagnetism, and Larmor’s radiation formula are not wrong in themselves—they remain excellent approximations for macroscopic charged objects. Yet, they fail for the electron in the hydrogen atom because they embed a specific ontology:

- Electrons are *point particles* with definite position and momentum at every instant.
- They move along *continuous trajectories* governed by deterministic force laws.
- They act as *classical sources* for the electromagnetic field, coupling through their charge and acceleration.

It is known nowadays that each of these assumptions is false. A model can therefore be *mathematically consistent*, *empirically adequate in its domain*, and *ontologically erroneous*. The classical hydrogen atom is a case study in exactly this combination.

2.2 The General Diagnostic Distinction

We propose the following general principle:

When a physical model fails empirically at the boundary of its domain, one must ask whether the failure is predictive or ontological. A predictive failure can often be corrected by modifying equations within the same ontological framework. An ontological failure requires replacing the picture of what exists.

As an example, the transition from Newtonian to relativistic mechanics is largely *predictive*: the ontology of point particles with definite trajectories is preserved; only force laws and the metric of spacetime change. The transition from classical to quantum mechanics is *ontological*: the very category of “particle with a definite trajectory” ceases to apply.

This distinction has implications for how we interpret theoretical progress. Kuhnian paradigm shifts [5] and Lakatosian research programmes [6] both capture aspects of it, but neither systematically separates the predictive from the ontological layer. The collapse calculation makes the separation vivid: the predictive machinery is flawless; it is the ontological presuppositions that must go.

2.3 A Second Case: The Ultraviolet Catastrophe

The atomic collapse is not an isolated case. The ultraviolet catastrophe [7, 8] presents the same diagnostic structure. Applying the equipartition theorem to the electromagnetic field modes in a cavity yields a spectral energy density $u(\nu, T) \propto \nu^2 k_B T$, which diverges as $\nu \rightarrow \infty$ —a prediction that is mathematically exact given classical statistical mechanics and electrodynamics.

Wien’s empirical law [9] and ultimately Planck’s quantum hypothesis [10] resolved the discrepancy not by correcting an equation within the classical framework but by replacing its ontological foundation: energy exchange between matter and field is not continuous but quantised in units $E = h\nu$. The failure, again, was not predictive within a domain but ontological at the domain boundary.

[TODO: Main point, choose direction. Just atom as case study, then generalize to others, or a main point at three levels in parallel.]

2.4 Current approaches and discuss bibliography

Cartwright’s argument in *How the Laws of Physics Lie* [11] is the closest precedent: she contends that the fundamental laws of physics achieve their generality by systematically misrepresenting causal capacities, and that models introduce *ad hoc* corrections that restore empirical accuracy at the cost of ontological transparency. Our claim is more specific: the classical atomic model does not merely misrepresent capacities—it presupposes a *category* of entity (the classically trajectoried particle) that simply does not exist at the relevant scale. The failure is not one of approximation but of kind.

Giere’s model-theoretic account [12] holds that models are abstract objects that *represent* target systems through a relation of similarity, with no claim to literal truth about underlying ontology. On this view, asking whether the classical atomic model “has the right ontology” is a category error: models represent, they do not describe. We resist this conclusion. The collapse calculation shows that the classical model does not merely *fail to represent* the hydrogen atom; it *entails* a prediction—sixteen picoseconds to collapse—that is straightforwardly and catastrophically false. That is not a similarity failure; it is an ontological commitment that nature rejects.

Chakravartty’s semi-realism [13] distinguishes *detection properties*—those causally responsible for our measurements—from *auxiliary properties* that models posit but that may not correspond to anything real. This is the closest structural analogue to our two-layer distinction. The difference is diagnostic emphasis: Chakravartty’s framework is designed to identify what to be realist *about* within a successful theory, whereas ours is designed to identify *why* a theory fails at a domain boundary. The classical collapse is a case where the auxiliary properties (definite position, continuous trajectory, classical field coupling) are not merely unconfirmed but actively contradicted. None of these three frameworks isolates the ontological layer as a separate *diagnostic target* at the moment of theoretical failure in quite the way we propose.

[TODO: Extend to other current authors addressing the issue. Find a real gap.]

3 Bohr's Provisional Solution and the Persistence of the Problem

3.1 The 1913 Postulates as Ontological Stipulation

Bohr's response in 1913 was philosophically unusual in its candour [14]. He did not attempt to repair the classical ontology. He declared classical electrodynamics inapplicable to intra-atomic orbits and introduced quantised stationary states as irreducible postulates: the electron in a stationary state does not radiate, not because some mechanism suppresses radiation, but because the very concept of radiating *simply does not apply to that state*.

This is an ontological statement, not a dynamical explanation. Bohr acknowledged as much, and indeed he described his postulates as provisional, awaiting a deeper theory. The immediate success was spectacular. The energy levels

$$E_n = -\frac{m_e e^4}{2\hbar^2} \cdot \frac{1}{n^2} = -\frac{13.6 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots \quad (11)$$

reproduce the Balmer series [15] and the full Rydberg formula [16], with the Rydberg constant now derived rather than fitted:

$$R_H = \frac{m_e e^4}{4\pi\hbar^3 c} \approx 1.097 \times 10^7 \text{ m}^{-1}, \quad (12)$$

a numerical agreement so precise it left no doubt that the quantisation condition was capturing something real, even without ontological justification.

The precision was stunning. Yet, the ontological justification was absent.

3.2 Quantum Mechanics: The problem remains

Full quantum mechanics relocates rather than dissolves the problem. In the Schrödinger picture, the ground-state electron occupies the state $\psi_{1s}(r) \propto e^{-r/a_0}$, spherically symmetric. $\psi(x)$ represents a probability density, hence there is no definite orbit, no classical centripetal acceleration, and therefore no Larmor radiation. Stability is explained: the ground state is the lowest eigenvalue of the Hamiltonian $\hat{H} = \hat{p}^2/2m_e - e^2/4\pi\epsilon_0 r$, and there is no lower state to decay into.

Quantum electrodynamics (QED) goes further, correctly computing excited-state lifetimes, the Lamb shift (~ 1057 MHz for the $2s-2p_{1/2}$ splitting, measured by Lamb and Retherford in 1947 [17]), and spontaneous emission rates at parts-per-billion precision.

Yet the problem has a residue. The self-energy of the electron in its own electromagnetic field has no fully satisfactory formulation. The Abraham–Lorentz equation [18, 19, 20],

$$m_e \dot{v} = F_{\text{ext}} + \frac{e^2}{6\pi\epsilon_0 c^3} \ddot{v}, \quad (13)$$

admits *runaway solutions* (acceleration growing exponentially without external force) and *pre-acceleration solutions* (particle responds before the force arrives, violating causality). In QED these pathologies are controlled by renormalisation, which works numerically with extraordinary precision but which Dirac regarded as a fundamental logical inconsistency [20] and Feynman called sweeping the problem under the rug with an elegant technique [21].

The ontological question Bohr implicitly raised in 1913—what *kind of thing* is an electron, considered as both a quantum object and a source of its own field?—remains incompletely answered. Each theoretical layer has provided a more adequate partial ontology. None has provided a complete one.

This situation is perhaps better understood through the lens of effective field theory [22]: rather than a deficiency awaiting resolution, ontological incompleteness at each energy scale is a permanent structural feature of physical theorising. Each theory carries a partial ontology adequate to its domain; what lies beyond is encoded in the theory's own breakdown, not in a completed picture waiting to be found.

[TODO: Extend paragraph on effective field theory (Weinberg, Wilson): EFTs explicitly embrace ontological incompleteness at each scale. That may be a more honest framing than “the problem is unsolved.”]

4 Aristotle’s Potentiality and the Quantum State

4.1 Heisenberg’s Invocation

In his famous *Physics and Philosophy* (1958), Heisenberg brilliantly introduces an Aristotelian category to interpret quantum mechanical phenomena [23]:

“[The probability wave] was a quantitative version of the old concept of *potentia* in Aristotelian philosophy. It introduced something standing in the middle between the idea of an event and the actual event, a strange kind of physical reality just in the middle between possibility and reality.”

This invocation is often treated as a philosophical metaphor appended to an otherwise self-contained physical theory. We argue it is more: it is a serious ontological proposal that deserves reconstruction and critical evaluation on its own terms.¹

[TODO: Two authors have already written about this and are reference for this topic, check them out.]

4.2 The Aristotelian Framework

Aristotle’s *Metaphysics* [25] distinguishes two fundamental modes of being:

- δύνανμις (potentiality): what a thing can be or do, considered as a real but not yet actualised property. Not mere logical possibility, nor ignorance about an already determinate state, but a genuine ontological category—a real property of things, prior to and distinct from their actual states.
- ἐνέργεια / ἐντελέχεια (actuality): the realisation of that potentiality; what a thing is in full exercise of its nature.

For Aristotle, bronze *is* potentially a statue before any sculptor acts; the seed *is* potentially a tree. These are not descriptions of what we happen not to know; they describe what the thing really is, in a mode of being distinct from full actuality.

4.3 A Note on Zubiri and the Etymology of Being

The recovery of Aristotelian potentiality for physics is not without precedent in the philosophical tradition, though it has gone largely unnoticed in the analytic philosophy of physics. Xavier Zubiri, in *Sobre la esencia* [26] and *Inteligencia sentiente* [27], developed a sustained critique of the ontological tradition precisely through attention to the etymology and original sense of the Greek *ὄν* (*on*, the present participle of *einai*, to be).

Zubiri’s central contention is that the Latin translation of *ὄν* as *ens*—and the subsequent scholastic identification of *esse* as an abstract noun—transformed what was in Aristotle a *dynamic* concept (being as an ongoing process, a *being-in-act*) into a *static* one (being as a fixed substantial nature). The classical ontology of the atom inherits precisely this Latin distortion: electrons are conceived as *entia* with fully determinate properties at every instant—a picture of being as complete, static actuality.

¹Barandes [24] shows that the full empirical content of quantum mechanics can be recovered from *indivisible stochastic processes*—transition probabilities satisfying a generalised Markov condition—without positing a wavefunction or Hilbert space as primitive ontology. Potentialities become transition probabilities between configurations, arguably closer to the Aristotelian notion of δύνανμις as process than the static wavefunction picture.

What Zubiri calls *la realidad* is not a fixed substrate of properties but a *de suyo*—a being-from-itself that is irreducibly open, whose determinations emerge in relation to other realities and cannot be read off from an intrinsic nature considered in isolation. This is structurally congruent with the potentiality reading: the electron’s δύνάμις is not a deficient mode of being awaiting completion, but a genuine and irreducible mode of *de suyo* that precedes and grounds any particular actualisation.

Zubiri’s framework has not been applied to the philosophy of quantum mechanics, to our knowledge. We suggest it deserves to be: his critique of the static Latin *esse* maps precisely onto the failure of the classical atomic ontology, and his recovery of dynamic ζῶν maps onto the potentiality reading we are defending. The convergence is not merely terminological—it reflects a shared diagnosis that the metaphysical presuppositions inherited from scholastic substance ontology are inadequate to describe entities whose mode of being is irreducibly open and relational.

[TODO: Esto puede ser mas que una nota, hay mucho que rascar. Encontrar la cita original de "quien ignora las lenguas está ciego. Una frase potente sobre dedicar una vida entera a traducir el Almagesto.]

4.4 Application to the Quantum State

The mapping is structurally precise. Before a position measurement on an electron in state ψ , the electron does not have a definite position we merely fail to know—this is what Bell’s theorem and its experimental confirmation rule out [28, 29, 30].

Rather, the electron has a distribution of *potential* positions, encoded by the so-called wave function—more accurately, it’s hermitian square, or probability— $|\psi(x)|^2$, which are real properties of the electron’s present state. Measurement does not reveal a pre-existing value; it *actualises* one of the potentialities.

On this reading:

- ψ is the mathematical representation of the electron’s potentialities.
- Measurement is the physical process of passage from potentiality to actuality.
- The Born rule $|\psi(x)|^2 dx$ gives the quantitative structure of the potentiality distribution.

This is not the Copenhagen interpretation in its instrumentalist form, which treats ψ as a bookkeeping device for predictions. It is a realist reading: the potentialities are real features of the world, not features of our knowledge.

[TODO: También podría ser al revés. Que la QM deba dejar la ontología de lado, y dedicarse a la epistemología, es una teoría de nuestra información sobre el sistema, no del sistema en si. Las probabilidades al final son eso.]

4.5 What This Resolves and What It Does Not

The Aristotelian reading dissolves the puzzle of the classical collapse. The $1s$ electron has no definite trajectory; there is no classical kinematic state to insert into Larmor’s formula (3). The question “why does the orbiting electron not radiate?” does not arise, because its presupposition—that the electron has a definite centripetal acceleration—is false.

What this reading does *not* resolve is the measurement problem in its general form. It relocates the hard question to the actualisation process: what determines which potentiality becomes actual? The major interpretations map partially onto different answers:

- **Copenhagen:** actualisation is a primitive, theory-external fact; quantum mechanics describes potentialities but not the process of actualisation itself.
- **GRW:** spontaneous collapse provides a physical mechanism for actualisation, at the cost of modifying the Schrödinger equation [31].
- **Everett:** all potentialities are actualised in branching worlds; no collapse occurs [32].
- **Bohmian mechanics:** particles always have definite positions (full actuality); the pilot wave guides them deterministically [33].

The measurement problem becomes sharper still in Wigner’s thought experiment of 1961 [34]. An observer inside an isolated laboratory—Wigner’s friend—measures the spin of an electron and, from his perspective, obtains a definite outcome. Wigner, outside the laboratory, describes the entire system (friend, apparatus, electron) as a closed quantum system evolving unitarily, assigning it the entangled state

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle_e|\text{saw } \uparrow\rangle_F + |\downarrow\rangle_e|\text{saw } \downarrow\rangle_F) . \quad (14)$$

The two observers assign incompatible quantum states to the same physical system at the same moment. On the potentiality reading, the paradox sharpens a question that our framework does not fully answer: is the passage from potentiality to actuality—from *δύναμις* to *ἐνέργεια*—an objective physical event, or is it indexed to a particular observer’s interaction with the system? If the former, when exactly does it occur? If the latter, the potentialities are not fully ontic in the sense we have been defending.

Frauchiger and Renner [35] sharpened this into a no-go theorem: assuming simultaneously that quantum mechanics is universal, that different observers’ reasoning is mutually consistent, and that the Born rule applies, leads to contradiction. The theorem does not refute any single interpretation but shows that the three assumptions cannot all be true together—each major interpretation must abandon at least one. For the potentiality realist, the theorem suggests that the ontology of actualisation requires further precision: a fully satisfactory account must specify the physical conditions under which joint potentialities become actual, and for whom.

We do not adjudicate between these here. Our claim is more modest: the Aristotelian framework provides a more adequate *starting* ontology for quantum mechanics than either classical particle realism or strict instrumentalism, and the collapse calculation is the sharpest way to see why.

[TODO: Reviewer will ask: is potentiality realism distinguishable from an epistemic interpretation? Key argument: potentialities are ontic, not epistemic—they ground the Born-rule probabilities rather than describing ignorance. [36], [37]]

A natural objection must be addressed directly: is the potentiality reading genuinely *ontic*, or is it merely a reformulation of an epistemic interpretation in Aristotelian vocabulary? The objection runs as follows—the wavefunction ψ describes our knowledge of the system; “potentiality” is just a philosophically dressed version of “probability given our ignorance”; and the Aristotelian language adds no ontological content.

We resist this reading on two grounds.

First, Bell’s theorem [28] and its experimental confirmation [29, 30] rule out any interpretation on which the wavefunction merely encodes ignorance about pre-existing local values. If $|\psi(x)|^2$ described our ignorance of a definite pre-existing position, the CHSH inequality would not be violated. It is violated. The potentialities encoded in ψ cannot therefore be epistemic in the relevant sense—they are not ignorance about something that is already actual.

Second, Margenau argued as early as 1954 [36] that quantum states describe *latencies*—real dispositional properties of physical systems that are not reducible to the statistics of measurement outcomes. Bunge [37] developed this into a systematic ontology of propensities, arguing that quantum probabilities are grounded in real physical tendencies of systems, not in subjective degrees of belief or objective frequencies over ensembles. On both accounts, the potentialities are *ontic*: they are features of the world that ground the Born-rule statistics, not descriptions of our epistemic state with respect to an already-actual world.

The potentiality reading we are defending is therefore not epistemic Copenhagen in disguise. It is a realist position that takes the wavefunction to represent real features of the world—features that are genuinely potential rather than actual, in the Aristotelian sense, prior to measurement. The difference from standard collapse interpretations is that we do not treat potentiality as a temporary state of ignorance to be resolved by measurement into something fully actual and classical. Potentiality is a permanent and irreducible mode of being for quantum systems between interactions—it is what they *are*, not what we *don't know* about them.

5 Entanglement, Structural Realism, and Conclusion

5.1 Entanglement as Irreducibly Relational Potentiality

The potentiality reading connects naturally to ontic structural realism (OSR) [38, 39]. OSR proposes that what is real are structures—patterns of relations—rather than objects with intrinsic natures, motivated in part by the permutation symmetry of identical particles [40] and by the non-locality revealed by Bell inequality violations [28, 29, 30].

Consider the spin-singlet state of two electrons,

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle_A |\downarrow\rangle_B - |\downarrow\rangle_A |\uparrow\rangle_B). \quad (15)$$

This state cannot be written as a product $|\phi\rangle_A \otimes |\chi\rangle_B$. Neither particle A nor B has a definite spin potentiality on its own. On the Aristotelian reading, the potentialities are properties not of A and B separately but of the *pair* AB jointly—an irreducibly relational structure.

Bell’s theorem [28] shows that no local hidden-variable theory can reproduce the quantum correlations. The CHSH inequality

$$|E(a, b) - E(a, b') + E(a', b) + E(a', b')| \leq 2 \quad (16)$$

is violated by quantum mechanics up to $2\sqrt{2}$ (the Tsirelson bound), confirmed experimentally [29, 30]. Entanglement is not a correlation between independently real objects. It is a structure of irreducibly joint potentiality with no local decomposition—precisely the structural realist’s claim that relations are ontologically prior to relata.

5.2 Potentiality Realism vs. Ontic Structural Realism

The question arises whether the Aristotelian potentiality reading simply *is* a form of OSR. We suggest it occupies a distinct position. OSR in its strongest form eliminates objects entirely in favour of structures [40]. The Aristotelian reading preserves objects—the electron is a genuine entity—but denies that objects are fully actual with respect to all their properties prior to measurement.

This is a form of *potentiality realism*: realism about both objects and their potentialities, with actuality restricted to what has been, or is being, measured. Objects are real; their potentialities are real; their actualities are contingent on measurement interactions.

[TODO: Engage with Esfeld and Deckert’s minimalist ontology [41]: points in a structure with no intrinsic properties. Argue potentiality realism is distinct: it accepts intrinsic properties *in potentia*, not *in actu*.]

The distinction from Esfeld and Deckert’s minimalist ontology [41] is worth making explicit. Their account posits structureless points whose only properties are relational—there are no intrinsic properties at all, even potential ones. Potentiality realism differs: it accepts that objects have intrinsic properties, but holds that those properties are possessed *in potentia* prior to measurement rather than *in actu*. The electron has a real dispositional tendency to be found in some region of space; that tendency is intrinsic to it even before any measurement actualises it. This is a genuinely distinct ontological position—neither the full structural elimination of Esfeld and Deckert, nor the classical picture of fully actual intrinsic properties at every instant.

5.3 Conclusion

The classical calculation of the hydrogen atom’s radiative collapse is more than a historical curiosity. It is a precise, mathematically impeccable derivation—from Newton, Coulomb, and Larmor, all independently confirmed in their domains—that produces an empirically false result. And the reason it is false is not a flaw in the equations. It is a flaw in the ontology they presuppose.

We have argued for three connected claims. First, every physical model carries an ontological layer analytically distinct from its predictive-mathematical content; diagnosing which layer has failed is a non-trivial philosophical task that physics alone cannot perform. Second, Bohr’s 1913 resolution was an explicit ontological stipulation—he declared the classical ontology inapplicable without fully replacing it—and the residue of this incompleteness persists in the renormalisation programme of QED. Third, Aristotle’s potentiality/actuality distinction, recovered by Heisenberg in his 1958 essay, provides a more adequate starting ontology for quantum mechanics than either classical particle realism or strict instrumentalism. Finally we discuss entanglement, which read through this lens, is a structure of irreducibly joint potentiality that connects naturally to ontic structural realism while preserving a genuine role for objects.

The lesson is general. Scientific progress is not only the accumulation of more accurate predictions. It is the successive replacement of ontologies—each adequate to a domain, each carrying implicit metaphysical commitments, each eventually confronted with phenomena that force not just new equations but a new picture of what there is.

The Wigner’s friend paradox, discussed in Section 4.5, marks the boundary of what the present account can deliver: a fully satisfactory potentiality realism must eventually specify the physical conditions under which actualisation occurs and for whom, a task we leave for further work.

Sixteen picoseconds. That is how long a classical hydrogen atom would survive. That it does not survive that way, and that we needed a century of physics and philosophy to understand why, says something important about the relationship between mathematics, experiment, and the picture of the world that any physical theory always—whether acknowledged or not—carries with it.

6 Appendix: Bell Inequalities and the Tsirelson Bound

This note collects the mathematical background behind the claims made in Sections 4 and 5 regarding Bell’s theorem and its consequences. It is intended as a self-contained reference for readers unfamiliar with the formalism.

1. The EPR argument and the problem of correlations

Einstein, Podolsky, and Rosen argued in 1935 [42] that if two spatially separated particles produce perfectly anti-correlated measurement outcomes without any possible causal connection between the measurements, then the outcome of each measurement must have been determined in advance by some property carried locally by each particle—a *hidden variable*. Quantum mechanics, they argued, is incomplete because it does not specify this variable.

Bell [28] converted this from a philosophical dispute into a precise mathematical question: what statistical correlations are compatible with *any* local hidden-variable theory, regardless of its internal mechanism?

2. Bell’s original inequality and Pearson correlations

Suppose two particles, A and B , are produced together and travel to separate detectors. Each detector can be oriented in one of several directions, and each measurement yields an outcome ± 1 . Let λ be a hidden variable (or set of variables) distributed according to some probability density $\rho(\lambda)$, unknown but fixed at the moment the particles are produced. The local hidden-variable assumption is that the outcome at A depends only on the local setting a and on λ —not on what B does—and symmetrically for B :

$$A = A(a, \lambda) \in \{-1, +1\}, \quad B = B(b, \lambda) \in \{-1, +1\}. \quad (17)$$

The correlation between the outcomes at settings a and b is then the *Pearson correlation coefficient* of the two ± 1 random variables A and B :

$$E(a, b) = \int \rho(\lambda) A(a, \lambda) B(b, \lambda) d\lambda. \quad (18)$$

This is identical to the standard Pearson product-moment correlation $r_{AB} = \langle AB \rangle / (\sigma_A \sigma_B)$, which here reduces to $\langle AB \rangle$ because $\sigma_A = \sigma_B = 1$ for ± 1 variables. $E(a, b) = +1$ means perfect agreement, $E(a, b) = -1$ means perfect anti-correlation, and $E(a, b) = 0$ means statistical independence.

Bell showed that for any three measurement directions a, b, c , the correlations defined by (18) must satisfy:

$$|E(a, b) - E(a, c)| \leq 1 + E(b, c). \quad (19)$$

The proof is pure arithmetic: since $A(b, \lambda)B(b, \lambda) = \pm 1$, one has $|A(a, \lambda)B(b, \lambda) - A(a, \lambda)B(c, \lambda)| \leq 1 - A(b, \lambda)B(b, \lambda)A(b, \lambda)B(c, \lambda)$ pointwise in λ , and integrating over $\rho(\lambda)$ yields (19). No assumption about quantum mechanics enters—the inequality holds for *any* distribution $\rho(\lambda)$ and any ± 1 -valued functions $A(a, \lambda), B(b, \lambda)$.

3. The CHSH inequality

Clauser, Horne, Shimony, and Holt (1969) [43] reformulated Bell’s result in a form better suited to real experiments, which suffer from detector inefficiencies and finite statistics. The four authors’ initials give the inequality its name: **CHSH**.

The setup uses two settings per detector— a and a' for particle A , and b and b' for particle B —yielding four correlation functions. The CHSH quantity S is defined as:

$$S = E(a, b) - E(a, b') + E(a', b) + E(a', b'). \quad (20)$$

The letter S stands for the *sum* (or *score*) of these four Pearson correlations, combined with a specific sign pattern chosen so that any local hidden-variable model is forced into an algebraic contradiction. The CHSH inequality states:

$$|S| \leq 2. \quad (21)$$

The proof is again arithmetic: since each $A(\cdot, \lambda), A'(\cdot, \lambda), B(\cdot, \lambda), B'(\cdot, \lambda) \in \{-1, +1\}$, the combination $AB - AB' + A'B + A'B' = A(B - B') + A'(B + B')$ takes values in which one of $(B - B')$ and $(B + B')$ is always zero and the other is ± 2 , so the whole expression is ± 2 pointwise in λ , and therefore $|S| \leq 2$ after integrating over $\rho(\lambda)$.

4. Quantum mechanics violates the CHSH bound

For the spin-singlet state $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle_A |\downarrow\rangle_B - |\downarrow\rangle_A |\uparrow\rangle_B)$, quantum mechanics predicts:

$$E_{\text{QM}}(a, b) = -\cos(\theta_{ab}), \quad (22)$$

where θ_{ab} is the angle between the measurement directions a and b in the Bloch sphere. Note that (22) is still a Pearson correlation between ± 1 outcomes—the quantum calculation gives the same object as (18), it just predicts a *different* value for it than any local hidden-variable model can.

Choosing the optimal angles $a = 0^\circ$, $a' = 90^\circ$, $b = 45^\circ$, $b' = 135^\circ$ (so that each pair of settings subtends 45°):

$$E(a, b) = E(a', b) = E(a', b') = -\cos 45^\circ = -\frac{1}{\sqrt{2}}, \quad E(a, b') = -\cos 135^\circ = +\frac{1}{\sqrt{2}}, \quad (23)$$

giving:

$$S_{\text{QM}} = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = -2\sqrt{2} \approx -2.828. \quad (24)$$

The magnitude $|S_{\text{QM}}| = 2\sqrt{2} > 2$ violates the CHSH bound (21). This violation was confirmed experimentally by Aspect, Grangier, and Roger in 1982 [29], and without loopholes by Hensen et al. in 2015 [30].

5. The Tsirelson bound: why not stronger?

The CHSH bound for classical correlations is $|S| \leq 2$. Quantum mechanics reaches $|S| = 2\sqrt{2} \approx 2.83$. But why does quantum mechanics stop there? Logically, $|S|$ could be as large as 4 (if all four Pearson correlations had magnitude 1 simultaneously). Is there a deeper reason why the quantum world is *more* non-local than any classical model but *less* non-local than it could logically be?

Boris Tsirelson (also transliterated Cirel'son) answered this in 1980 [44]: $2\sqrt{2}$ is the maximum value of $|S|$ achievable by any quantum state and any quantum measurements, in any Hilbert space of any dimension. This is the **Tsirelson bound**.

To see intuitively why it is $2\sqrt{2}$ and not 4: the four Pearson correlations $E(a, b)$, $E(a, b')$, $E(a', b)$, $E(a', b')$ cannot all equal ± 1 simultaneously in quantum mechanics, because the measurements do not commute. Choosing a and b to maximise $E(a, b)$ forces the other correlations into a configuration that cannot simultaneously be maximised—the non-commutativity of quantum observables introduces a geometric constraint that limits the total score S .

The algebraic proof uses the operator version of S :

$$\hat{S} = \hat{A} \otimes (\hat{B} + \hat{B}') + \hat{A}' \otimes (\hat{B} - \hat{B}'), \quad (25)$$

where $\hat{A}, \hat{A}', \hat{B}, \hat{B}'$ are observables with eigenvalues ± 1 (Hermitian operators with $\hat{A}^2 = \mathbb{I}$, etc.). Computing $\hat{S}^2$ and using $\hat{A}^2 = \hat{A}'^2 = \hat{B}^2 = \hat{B}'^2 = \mathbb{I}$:

$$\hat{S}^2 = 4\mathbb{I} + [\hat{A}, \hat{A}'] \otimes [\hat{B}, \hat{B}']. \quad (26)$$

The operator norm of a commutator $[\hat{X}, \hat{Y}]$ for ± 1 observables satisfies $\|[\hat{X}, \hat{Y}]\| \leq 2$, so $\|\hat{S}^2\| \leq 4 + 4 = 8$, giving $\|\hat{S}\| \leq 2\sqrt{2}$. The bound is tight: the singlet state with 450 settings achieves it exactly.

6. The intermediate bound and its philosophical meaning

The existence of an intermediate bound—classical ≤ 2 , quantum $\leq 2\sqrt{2}$, logical maximum $= 4$ —is not an accident, and its philosophical significance goes beyond confirming quantum mechanics.

Popescu and Rohrlich noted in 1994 [45] that one can define hypothetical theories, now called *PR-boxes*, that respect the no-signalling principle (no faster-than-light communication) but saturate $|S| = 4$. Such theories are logically consistent and non-signalling, yet they would allow correlations far stronger than anything quantum mechanics predicts. The fact that nature chooses $2\sqrt{2}$, and not 4, is therefore a *non-trivial physical fact* that is not explained by no-signalling alone. Why this particular number?

Several research programmes have tried to derive $2\sqrt{2}$ from information-theoretic principles—non-trivial communication complexity [46], information causality [47], and local orthogonality [48]—but no single principle has been universally accepted as the explanation.

For our purposes, this is directly relevant to the potentiality reading defended in Section 5. The Tsirelson bound shows that the joint potentialities encoded in the quantum state are not arbitrary: they obey a precise algebraic structure (the Hilbert space formalism, and specifically the constraint (26) on operator norms) that limits *how strongly* the potentialities of two entangled systems can be correlated.

The entanglement is not a free, unlimited non-local resource—it is a structured form of joint potentiality, constrained by the algebra of quantum observables. This is consistent with potentiality realism: the potentialities are *real* (they exceed classical bounds), but they are *structured* (they obey the Tsirelson bound rather than the logical maximum).

The gap between $2\sqrt{2}$ and 4 is, in this sense, a quantitative signature of the ontological structure of quantum potentiality—a structure that neither classical ontology nor pure logic would have predicted, and that remains only partially understood.

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